

Math Bootcamp: Content to be Covered

For Incoming MSc Economics Students

Math Camp 2026

Overview

- **Objective:** Bridge the gap between a weaker undergraduate math background and MSc-level economics courses.
- **Duration:** 3 hours per day for 5 days.
- **Method:** Focus on intuition, fundamentals, and confidence-building. Rigorous proofs and advanced techniques are deferred to fall courses; only simple proofs are included where necessary.
- **Emphasis:** Conceptual topics to be covered throughout graduate microeconomics, macroeconomics, econometrics, and mathematical economics.

Contents

1	Day 1 — Foundations (Sections 1.1–1.9)	3
1.1	1.1 — Notation and Number Systems	3
1.2	1.2 — Logic and Mathematical Statements	3
1.3	1.3 — Sets and Set Operations	3
1.4	1.4 — Functions	3
1.5	1.5 — Supremum and Infimum	3
1.6	1.6 — Algebra Revision	4
1.7	1.7 — Single-Variable Derivatives	4
1.8	1.8 — Multivariate Calculus	4
1.9	1.9 — Limits	4
1.10	Appendix (Slides)	5
2	Day 2 — More Calculus and Real Analysis Bridge	5
2.1	2.1 — Implicit Function Theorem	5
2.2	2.2 — Mean Value Theorem	5
2.3	2.3 — Integration	5
2.4	2.4 — Leibniz Rule and Fubini	6
2.5	2.5 — Taylor Expansion and Linearization	6
2.6	2.6 — Real Analysis Bridge (must cover, Block 3)	6
2.7	Appendix (Day 2 Slides)	7
3	Day 3 — Linear Algebra and Real Analysis	7
3.1	Linear Algebra — must cover (Block 1)	7
3.2	Convergence — must cover (Block 2)	8
3.3	Real Analysis — if time allows (Block 3)	8

4	Day 4 — Optimization	8
4.1	Convex Sets and Convex Functions	8
4.2	Quasiconcavity and Quasiconvexity	8
4.3	Unconstrained Optimization	9
4.4	Constrained Optimization: Equality Constraints	9
4.5	Constrained Optimization: Inequality Constraints	9
4.6	Envelope Theorem	9
5	Day 5 — Dynamic Programming	9
5.1	Motivation and Framework	10
5.2	State Variables versus Control Variables	10
5.3	Bellman Equation	10
5.4	Backward Induction	10
5.5	Value Function Iteration (Conceptual)	10
5.6	Two-Period Models as Building Blocks	10
6	Cross-Cutting Themes (All Days)	11
7	Suggested Time Allocation	11

1 Day 1 — Foundations (Sections 1.1–1.9)

Approximate duration: 3 hours. See Day 1 slides (Sections 1.1–1.9).

1.1 1.1 — Notation and Number Systems

- Number systems: $\mathbb{N}$ (counting), $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{R}^n$; nesting $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.
- Symbols: $\in$, $\subset$, $\forall$, $\exists$, $\Rightarrow$, $\Leftrightarrow$.
- Scalars versus vectors: $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$; dot product $\mathbf{p} \cdot \mathbf{x} = \sum_i p_i x_i$.
- Practice: identify the smallest set containing a given object (e.g. counts in $\mathbb{N}$; prices in $\mathbb{Q}$; order vectors in product sets).

1.2 1.2 — Logic and Mathematical Statements

- P **only if** Q ($P \Rightarrow Q$): Q necessary for P .
- P **if** Q ($Q \Rightarrow P$): Q sufficient for P .
- P **iff** Q ($P \Leftrightarrow Q$): necessary and sufficient.
- Contrapositive: $\neg Q \Rightarrow \neg P$ equivalent to $P \Rightarrow Q$.
- Running example: IKEA budget set $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq B\}$; practice with $\in$, $\subset$, $\forall$, $\exists$, $\Rightarrow$, and contrapositive.

1.3 1.3 — Sets and Set Operations

- Notation: $\in$, $\notin$, $\subset$, $=$, $\emptyset$, roster and set-builder forms.
- Union, intersection, complement, difference; Cartesian product $A \times B$.
- Economic sets: budget sets, consumption sets; product spaces such as $\mathbb{Q} \times \mathbb{N}$ versus constraint sets like $\mathcal{B}$.
- Practice: describe $\mathcal{B} \cap \mathcal{H}$, $\mathcal{B} \cup \mathcal{H}$, $\mathcal{B} \setminus \mathcal{H}$; distinguish elements from subsets.

1.4 1.4 — Functions

- Function $f: A \rightarrow B$: domain, codomain, range $f(A) \subseteq B$.
- Injective (one-to-one) and surjective (onto); codomain versus range.
- Operations: $(f \pm g)$, (cf) , (fg) , composition $(f \circ g)(x) = f(g(x))$.
- A rule must assign **exactly one** output per input (not a correspondence).
- Practice: domain, codomain, range, injectivity, and surjectivity (e.g. partner age as a function of your age).

1.5 1.5 — Supremum and Infimum

- Example first: reservation price / willingness to pay when the purchase set $P = \{p \in \mathbb{R}_+ : p < \bar{p}\}$ has no maximum but $\sup P = \bar{p}$.
- $\max S$ / $\min S$ versus $\sup S$ / $\inf S$ (least upper bound, greatest lower bound).
- When extrema are attained versus only approached (e.g. $S = [0, 1)$, $\sup S = 1 \notin S$).
- If $\max S$ exists, $\max S = \sup S$; always $\inf S \leq x \leq \sup S$ for $x \in S$.

1.6 1.6 — Algebra Revision

- Taylor expansion around a : $f(a+h) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} h^k$ (first-order: $f(a+h) \approx f(a) + f'(a)h$).
- Log and exponential rules; series $\ln(1+x)$ and e^x .
- **Why logs:** $\Delta \ln x \approx \Delta x/x$ tracks percentage changes; elasticity $\varepsilon_{y,x} = \partial \ln y / \partial \ln x$; log-log regression.
- Cobb–Douglas: $U = x_1^\alpha x_2^{1-\alpha}$, $Y = AK^\alpha L^{1-\alpha}$; linear in logs; constant elasticities and expenditure shares.
- Local log approximation of any smooth $u > 0$ near $\bar{\mathbf{x}}$; Cobb–Douglas as a modeling choice when used globally.
- Homogeneity of degree k : $f(t\mathbf{x}) = t^k f(\mathbf{x})$; Cobb–Douglas utility and production are degree 1 (CRS: double inputs $\Rightarrow$ double output).

1.7 1.7 — Single-Variable Derivatives

- Formal definition: $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$; meaning as slope and marginal rate (MC , MU , MP_K).
- Rules: constants, sums, power, product, quotient, chain rule; $(e^x)' = e^x$, $(\ln x)' = 1/x$, $(a^x)' = a^x \ln a$; inverse-function rule (stated).
- Practice: cubic cost, monopolist profit, Cobb–Douglas $Y(K)$, composite cost (chain rule).
- **Euler’s theorem** (for C^1 homogeneous f of degree k): $\sum_i x_i f_{x_i} = k f(\mathbf{x})$ (proof via $t \mapsto f(t\mathbf{x})$).
- Adding-up problem: under CRS and competitive factor markets, $PQ = rK + wL$; IRS ($k > 1$) and DRS ($k < 1$) and payment exhaustion.

1.8 1.8 — Multivariate Calculus

- Level curves $\mathcal{L}_c = \{(x, y) : f(x, y) = c\}$ and contour maps; 3D slice picture for $z = f(x, y)$.
- Roadmap: partials $\Rightarrow$ total differential $\Rightarrow$ directional derivative $\Rightarrow$ cross-partial (Schwarz).
- Partial derivatives (limit definition; fix one variable); $MP_K = \partial Y / \partial K$ with L fixed.
- Total differential: add/subtract trick, MVT on each bracket, $dz = f_x dx + f_y dy$; Cobb–Douglas output example $dY = MP_K dK + MP_L dL$.
- Gradient ∇f : perpendicular to level curves; points uphill; $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cos \theta$.
- Directional derivative along unit direction $\mathbf{u}$; example: subsidy requiring input changes in ratio 2:1.
- Cross-partial and Schwarz’s theorem (stated); Y_{KL} and complements versus substitutes.

1.9 1.9 — Limits

- Motivation: Berkeley’s critique of “infinitely small” h in early calculus.
- Sequence limit (ε – N): quantifier order and convergence.
- Function limit (ε – δ): $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$; example $f(x) = x + 1$ for $x \neq 1$, $f(1) = 3$, $\lim_{x \rightarrow 1} f(x) = 2 \neq f(1)$.

- Continuity at a : $\lim_{x \rightarrow a} f(x) = f(a)$; differentiability at a : derivative limit exists.
- Differentiable $\Rightarrow$ continuous; converse fails ($|x|$ at 0; Weierstrass function).
- Practice: ε - δ proofs for $f'(a) = 2a$ when $f(x) = x^2$ and for $\lim_{x \rightarrow 1} f(x) = 2$.

1.10 Appendix (Slides)

- Derivations of single-variable rules: constants, sums, powers, product, quotient.
- Key limits for $(e^x)'$ and $(\ln x)'$; exponential and logarithm derivative proofs.

2 Day 2 — More Calculus and Real Analysis Bridge

Approximate duration: 3 hours. See Day 2 slides (Sections 2.1–2.6).

2.1 2.1 — Implicit Function Theorem

- Explicit versus implicit relationships; when $F(x, y) = 0$ cannot be solved in closed form.
- Comparative statics: $F(x, \theta) = 0$ defines $y^*(\theta)$; goal is $dy^*d\theta$.
- **Core recipe:** totally differentiate $F(x, y) = 0$ to get $dydx = -F_x F_y$ when $F_y \neq 0$.
- Regularity: $F_y \neq 0$ means finite slope locally; vertical tangent when $F_y = 0$.
- **Practice (Together):** nonlinear demand $D(p, \alpha) = \alpha e^{-p}$, supply $S(p) = p$; no closed form for $p^*(\alpha)$, but IFT still gives $dp^*d\alpha$.
- **Applications:** market clearing $E(p, \alpha) = D(p, \alpha) - S(p) = 0$; linear example checks IFT against explicit $p^*(\alpha) = \alpha - cb + d$.
- Direct versus indirect effects; chain-rule derivation of $dp^*d\alpha = -E_\alpha E_p$.
- **Tax incidence:** per-unit tax on sellers; $dp^*dt = db + d$; tax borne more heavily by the **less elastic** side.
- **Appendix (for interest):** several-variable / Jacobian form of IFT.

2.2 2.2 — EVT, MVT, and Existence

- **Open versus closed sets** (budget, strict budget, reservation price examples).
- EVT on closed interval $[a, b]$: continuous f attains max and min.
- When EVT fails: not closed, not bounded, or not continuous (failure figure).
- **Compact** in $\mathbb{R}^n$ (camp level): closed and bounded (Heine–Borel, stated).
- **Weierstrass EVT:** continuous f on compact K attains max and min.
- **Bridge:** compact feasible set + continuous objective $\Rightarrow$ optimum exists.
- Rolle's theorem; MVT; labor/output economics example.
- **Appendix (for interest):** Weierstrass EVT proof idea.

2.3 2.3 — Integration

- Riemann integral as signed area / accumulation; definite versus indefinite integrals.
- **For interest:** Lebesgue integral (measures level sets; handles sparse jumps).
- Fundamental Theorem of Calculus: $F'(x) = f(x) \Rightarrow \int_a^b f(x) dx = F(b) - F(a)$.
- Integration by parts: $\int u dv = uv - \int v du$ (product-rule proof); example $\int x e^x dx$.
- **Probability link:** PMF and CMF (discrete dice); PDF and CDF (continuous $\text{Unif}(0, 1)$); FTC link $F'(x) = f(x)$.
- Expectation for non-negative continuous X : $\mathbb{E}[X] = \int_0^\infty x f(x) dx = \int_0^\infty (1 - F(x)) dx$.
- **Appendix (for interest):** expectation-from-CDF proof (integration by parts).

2.4 2.4 — Leibniz Rule and Fubini

- **Fixed limits:** if $F(\theta) = \int_a^b f(x, \theta) dx$, then under regularity $dF d\theta = \int_a^b \partial f / \partial \theta(x, \theta) dx$.
- Regularity (camp level): $\partial f / \partial \theta$ exists and $f, \partial f / \partial \theta$ are continuous on $[a, b] \times I$.
- **Variable limits:** when $a(\theta), b(\theta)$ move, $F'(\theta)$ adds endpoint terms plus the integral of $\partial f / \partial \theta$.
- When fixed-limit Leibniz fails: moving discontinuity (step function) example.
- **Fubini:** if $\int_D |f(x, y)| dx dy < \infty$, then $\int_D f = \int \int f$ (order of integration can be swapped).
- Camp shortcut: bounded continuous f on a nice region; Tonelli (for interest) when $f \geq 0$.
- Example: $\int_D (x + y) dx dy$ on triangle $D = \{0 \leq x \leq 1, 0 \leq y \leq x\}$.

2.5 2.5 — Taylor Expansion and Linearization

- One-variable Taylor through order n ; multivariable C^2 form with ∇f and Hessian H_f .
- First-order truncation: $f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h}$.
- Second-order truncation: adds $\frac{1}{2} \mathbf{h}^\top H_f(\mathbf{a}) \mathbf{h}$.
- Two-variable explicit formula from matrix multiplication (optional if Hessian is familiar).
- When linearization works: small shifts, modest curvature, differentiability; fails with large shifts, kinks, strong nonlinearity.
- **Practice (Together):** first-order two-variable Taylor for $f(x, y) = x^2 + y$ at $(1, 1)$.
- **Take home:** second-order two-variable Taylor (e.g. $f(x, y) = \sin x + \cos y$ at $(0, 0)$).
- **Appendix (for interest):** one-variable Taylor proof; Itô's lemma and Brownian motion.

2.6 2.6 — Uniform Convergence

- Pointwise convergence: $f_n(x) \rightarrow f(x)$ for each fixed x .
- Uniform convergence: one N works for all x in the domain.

- Example: $f_n(x) = x^n$ on $[0, 1]$ converges pointwise but not uniformly.
- **Why it matters:** uniform convergence justifies swapping limits with continuity, integration, and (with extra conditions) differentiation.
- Link to section 2.4: Leibniz/Fubini need regularity so limits and integrals can be interchanged safely.

2.7 Appendix (Day 2 Slides)

- IFT — several variables (Jacobian form).
- Weierstrass EVT — proof idea.
- Expectation from the CDF (integration by parts).
- Leibniz rule failure (moving step).
- Taylor expansion proof (one variable); Itô's lemma and Brownian motion (for interest).

3 Day 3 — Linear Algebra and Real Analysis

*Approximate duration: 3 hours. Linear algebra is **must cover**; real-analysis topics below are prioritized.*

3.1 Linear Algebra — must cover (Block 1)

- **Vectors in $\mathbb{R}^n$:** notation $\mathbf{x} = (x_1, \dots, x_n)$; dot product $\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$ (expenditure $\mathbf{p} \cdot \mathbf{x}$, projections).
- **Matrices:** dimensions $m \times n$, entries, matrix–vector product $A\mathbf{x}$; addition, scalar multiplication, matrix multiplication (order matters).
- **Transpose** $A^\top$; identity matrix I .
- **Linear systems:** $A\mathbf{x} = \mathbf{b}$; solution as intersection of hyperplanes (economic: market clearing, input–output balance).
- **Determinant** (must know 2×2 and 3×3 formulas): $\det A \neq 0 \Leftrightarrow A$ invertible (square case).
- **Inverse** A^{-1} : $AA^{-1} = I$; solve $\mathbf{x} = A^{-1}\mathbf{b}$ when it exists.
- **Linear independence** (intuitive): no vector in a set is a linear combination of the others; dependence when one column is redundant.
- **Rank** (conceptual): number of linearly independent rows/columns; full rank $\Leftrightarrow$ invertible for square A .
- **Jacobian matrix:** $J_f(\mathbf{x}) = (\partial f_i / \partial x_j)$; local linear map of a multivariable function; link to Day 2 IFT appendix.
- **Quadratic forms** $\mathbf{x}^\top A \mathbf{x}$ (stated only): positive/negative definiteness and curvature of f near a critical point (preview Day 4 SOC).
- **IFT in matrix form:** if $\mathbf{F}(\mathbf{y}, \theta) = \mathbf{0}$ and $\partial \mathbf{F} / \partial \mathbf{y}$ is invertible, then $\partial \mathbf{y}^* / \partial \theta = -(\partial \mathbf{F} / \partial \mathbf{y})^{-1} (\partial \mathbf{F} / \partial \theta)$.

3.2 Convergence — if time allows (Day 3)

- Deeper practice with uniform convergence (Day 2.6 is the core statement).
- Additional counterexamples and exchange-of-limit applications.

3.3 Real Analysis — if time allows (Day 3)

- **Separation theorem** (picture-based): disjoint convex sets separated by a hyperplane; supporting hyperplanes; welfare economics and Pareto efficiency (high level).
- **Fixed point theorems** (statement): equilibrium as a fixed point; Brouwer or contraction mapping; existence of equilibrium (no proof).
- **Continuity recap**: ε - δ at a point versus on a set; continuity of demand, supply, profit functions; when discontinuities break optimization.
- **Product spaces** $\mathbb{R}^n$: choice sets as subsets of product domains (brief, if not done Day 1).
- **Metric spaces / neighborhoods** (for interest only): distance, open balls; optional rigor beyond open/closed sets from Day 2.

4 Day 4 — Optimization

Approximate duration: 3 hours.

4.1 Convex Sets and Convex Functions

- Definition and geometry of convex sets.
- Convex and concave functions.
- Upper and lower contour sets (level sets).
- Why convexity or concavity guarantees global optima from first-order conditions.

4.2 Quasiconcavity and Quasiconvexity

- Definition of quasiconcave and quasiconvex functions via contour sets.
- Relationship between concavity/convexity and quasiconcavity/quasiconvexity.
- Why quasiconcavity is the standard regularity condition in utility maximization.
- When first-order conditions are sufficient for a global optimum under quasiconcavity.
- Strict quasiconcavity and uniqueness of optima (conceptual).
- Invariance under monotone transformations.
- Distinction between quasiconcavity of the objective and convexity of the feasible set in constrained problems.

4.3 Unconstrained Optimization

- Local versus global optima.
- First-order conditions: necessary conditions for interior optima.
- Second-order conditions: curvature and nature of critical points.

- Concavity and convexity of objective functions and uniqueness of optima.
- Economic optimization problems as a class (profit, utility, cost minimization).

4.4 Constrained Optimization: Equality Constraints

- Feasible set defined by equality constraints.
- Lagrange multiplier method: setup and interpretation.
- Lagrange multipliers as shadow prices or marginal values of constraints.
- Necessary conditions for constrained optima (Lagrange theorem, stated).
- Utility maximization subject to budget constraint (as problem type).

4.5 Constrained Optimization: Inequality Constraints

- Feasible set with inequalities.
- Binding versus non-binding constraints.
- Complementary slackness (conceptual).
- Karush–Kuhn–Tucker (KKT) conditions (statement and interpretation).
- Nonnegativity constraints and corner solutions.

4.6 Envelope Theorem

- **Placement:** after unconstrained and constrained optimization (FOCs, Lagrange, KKT).
- Value function of a parameterized problem: $V(\theta) = \max_x f(x; \theta)$ subject to constraints.
- **Unconstrained envelope theorem:** $\frac{dV}{d\theta} = \frac{\partial f}{\partial \theta} \Big|_{x=x^*(\theta)}$ (no need to differentiate the optimizer $x^*(\theta)$ directly).
- **Constrained envelope theorem:** $\frac{dV}{d\theta} = \frac{\partial \mathcal{L}}{\partial \theta} \Big|_{(x^*, \lambda^*)}$ with binding multipliers capturing constraint movements.
- Interpretation: shadow prices and comparative statics from the optimized value.
- Economic examples: profit as a function of a cost shifter or output price; indirect utility as a function of income and prices; expenditure/cost functions and Shephard-type logic (conceptual).
- Contrast with naive chain rule through $x^*(\theta)$: envelope theorem drops terms that vanish at the optimum.

5 Day 5 — Dynamic Programming

Approximate duration: 3 hours.

5.1 Motivation and Framework

- Sequential decision-making over time.
- Intertemporal trade-offs.
- Finite versus infinite horizon (finite focus for bootcamp).

- Markov structure: current state summarizes relevant history.

5.2 State Variables versus Control Variables

- State: variables that evolve according to law of motion.
- Control: variables chosen each period.
- Transition equation or law of motion.
- Feasibility constraints in dynamic problems.
- Classic problem types: consumption–savings, capital accumulation, resource extraction.

5.3 Bellman Equation

- Value function: maximum attainable payoff from a given state.
- Principle of optimality (Bellman’s idea).
- Recursive formulation of dynamic problems.
- Bellman equation for finite-horizon problems.
- Terminal condition and backward solution logic.

5.4 Backward Induction

- Solving finite-horizon problems from the last period backward.
- Relationship between period-by-period optimization and Bellman recursion.
- Policy function: mapping from state to optimal control.

5.5 Value Function Iteration (Conceptual)

- Idea of iterating on the Bellman operator.
- Convergence to fixed point of Bellman equation (intuition only, no algorithm).
- Distinction between solving finite-horizon and infinite-horizon problems.

5.6 Two-Period Models as Building Blocks

- Structure of two-period consumption–savings problems.
- Euler equation (conceptual introduction if appropriate).
- Link between dynamic programming and standard intertemporal optimization.

6 Cross-Cutting Themes (All Days)

- Mathematical notation used in economics graduate courses.
- Reading and interpreting first-order conditions, second-order conditions, and comparative statics.
- Connecting mathematical objects to economic meaning (marginal, shadow price, value function).
- Confidence with standard problem templates used in fall courses.

7 Suggested Time Allocation

Day	Block 1 (~1 hr)	Block 2 (~1 hr)	Block 3 (~1 hr)
1	1.1–1.4: notation; logic; sets; functions	1.5–1.6: sup/inf; algebra; Cobb–Douglas	1.7–1.9: derivatives; Euler; multivariate; limits
2	2.1 IFT	2.2 EVT/MVT; 2.3 integration	2.4–2.5; 2.6 uniform conv.
3	Linear algebra (must)	Convergence practice (if time)	Separation; fixed points (if time)
4	Convexity; quasiconcavity	Unconstrained optimization	Lagrange; KKT; envelope theorem
5	Framework; state and control	Bellman equation	Backward induction; VFI idea